Hot sauce and chili sauces: HMTc Standard

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CITE AS

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified standard for Hot sauce and chili sauces (Herbs, Spices, Seasonings and Salt), stated as consumed. Each analyte carries the strictest applicable government maximum level in the product's native basis, with a disclosed read-across or a mandatory control where no government regulates the exact product.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; explicit-id lookup; legislated product-form splits preserved; dried-spice basis discipline)

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	1,000	sovereign maximum level
Cadmium (Cd)	Tier 1	1,000	sovereign maximum level
Total arsenic (tAs)	Tier 2	1,000	sovereign maximum level
Inorganic arsenic (iAs)	Tier 1	100	sovereign maximum level
Mercury, total (tHg)	Tier 1	50	sovereign maximum level
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	—	—

Tin (Sn)	Tier 2	n/m	not material in this matrix
Aluminium (Al)	Tier 2	—	—
Total chromium (Cr)	Tier 2	—	—
Hexavalent chromium (Cr-VI)	Tier 2	—	—

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	1,000	government maximum level
Cadmium (Cd)	1,000	government maximum level
Total arsenic (tAs)	1,000	government maximum level
Inorganic arsenic (iAs)	100	government maximum level
Mercury, total (tHg)	50	government maximum level
Methylmercury (MeHg)	n/m	Methylmercury is the organomercury speciation of mercury and is arithmetically bounded by total mercury (MeHg <= tHg); a chili/vinegar/water sauce carries no mercury source and no methylmercury formation pathway, and it is bounded by the total-mercury sauce limit.
Nickel (Ni)	—	—
Tin (Sn)	n/m	Hot/chili sauces are sold in glass or plastic bottles, not tinplate cans, so there is no tin source or migration route. Material only for tin-plated canned product.
Aluminium (Al)	—	—
Total chromium (Cr)	—	—
Hexavalent chromium (Cr-VI)	—	—

4 | Cite this standard

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STATUS AND LICENCE

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COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://-heavymetalcertified.com/standards/hot-chili-sauces. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/hot-chili-sauces.